

Praveen Samarasekara

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🐙 github.com/RavPraveen

🌐 RavPraveenPortfolio.com

Summary

A passionate undergraduate with a strong interest in artificial intelligence, signal processing, and embedded systems. Always eager to explore new technologies and solve real-world problems through intelligent systems. I am actively seeking opportunities to apply my skills in research, industry, or project-based environments.

Education

Bachelor of Science (Hons) in Engineering — *University of Moratuwa*
Specializing in Electronic & Telecommunication Engineering

Jan 2024 – Present

- CGPA: 3.59 / 4.0 (Dean's List — Semester 1, Semester 4)

G.C.E. Advanced Level — *Bandaranayake Central College Veyangoda*
Physical Science Stream

2020

- 3A passes Z-Score: 2.5003 District Rank: 23

Projects

Customer Churn Prediction using Machine Learning(Ongoing)

🐙 GitHub

- **Tools:** Python, Scikit-learn, Pandas, NumPy, Matplotlib, Jupyter Notebook
- Developed a machine learning model to predict customer churn using a publicly available Kaggle dataset and performed comprehensive data preprocessing, feature engineering, and exploratory data analysis.
- Trained and evaluated a Random Forest classifier, achieving strong predictive performance through hyperparameter tuning and cross-validation.
- Analyzed feature importance to identify the key factors influencing customer retention and presented insights using data visualizations.

PrintEase Solutions – Automated Self-Service Printing Kiosk (*Frontend and Circuit design*)

🐙 GitHub

- **Tools:** React, Django, PostgreSQL, ESP32, PayHere API, Vite
- Developed an automated self-service printing kiosk that enables users to upload documents, complete secure online payments, and print without operator assistance.
- Integrated a React frontend, Django backend, and ESP32-based hardware interface to automate the document printing workflow and improve customer convenience.

Music Genre Classification Engine using PySpark MLlib (*Ongoing*)

- **Tools:** Python, Apache Spark, PySpark MLlib, Flask, Docker, HTML, CSS, JavaScript
- Engineered a scalable end-to-end Big Data machine learning pipeline using PySpark to classify music genres from song lyrics, covering data preprocessing, feature engineering, model training, and evaluation.
- Implemented a multi-model voting ensemble combining Gradient-Boosted Trees (One-vs-Rest), Naive Bayes, and Logistic Regression to improve classification performance.
- Developed a Flask web application with a REST API to serve the trained ensemble model, enabling real-time genre prediction through an interactive web interface.
- Containerized the complete application, including the Apache Spark environment and all dependencies, using Docker for reproducible, one-command deployment.

University Lost and Found System

🐙 GitHub

- **Tools:** Java, Node.js, npm
- Built a full-stack platform for reporting and reclaiming lost items on campus, with a Java backend and a Node.js/npm-based frontend.
- Worked on the REST API layer as well as the frontend, connecting the client to backend services for reporting, searching, and claiming lost items.

Multichannel Temperature Data Logger for Vaccine Cold Chain Transport

- **Tools:** Embedded C, RTC, SD Card Module, Wireless Communication Module, Microcontroller, Alert Interfaces
- Designed and developed a multichannel temperature data logger for monitoring vaccine cold chain transport, using a Human-Centered Design (HCD) approach.
- Implemented real-time clock (RTC) timestamping, SD card data storage, and an alert interface to flag temperature excursions across multiple channels.
- Added wireless connectivity to transmit temperature data and alerts remotely, enabling real-time monitoring during transport.

Tri-Mode Class AB Power Amplifier

- **Tools:** Analog Circuit Design, LTspice, Altium Designer, Power Electronics
- Designed and built an industrial-grade Class AB audio power amplifier using discrete analog components with support for Mono, Stereo, and Bridge operating modes.
- Developed the complete analog signal chain, including the power stage, biasing network, thermal stability, and protection circuitry to achieve high-fidelity audio performance.

Technical Skills

Languages: C, C++, Python, MATLAB, Java, JavaScript

Embedded / Hardware: ESP32, Arduino, Raspberry Pi

Frameworks & Tools: TensorFlow, PyTorch, OpenCV, NumPy, Git, GitHub, Docker

Software: MATLAB, Simulink, Altium, SolidWorks, VS Code

Operating Systems: Linux, Windows

Markup: LaTeX, Markdown, HTML, CSS

Honors & Awards

- Dean's List Placement — Semesters 1 — University of Moratuwa 2024, 2025
- Merit Pass — G.C.E. Advanced Level Examination 2020

Competitions & Achievements

- **Participant** — Sri Lanka Robotics Challenge (SLRC) 2025: Contributed to the design, development, and testing of an autonomous mobile robot for robotics competition tasks.

Volunteering

Committee Member — *Classical Music Society, University of Moratuwa* 2025 – Present

- Serve as a committee member, contributing to the planning and organization of concerts, workshops, and society events.
- Perform as a violinist in society concerts and musical events, promoting classical music within the university community.

Editor — *E-Carrier Annual Magazine, Department of Electronic and Telecommunication Engineering, University of Moratuwa* 2025 – Present

- Served as an editor for the department's annual magazine, reviewing, editing, and organizing technical and non-technical content for publication.

Courses & Certifications

- **Supervised Machine Learning: Regression and Classification** — *DeepLearning.AI & Stanford Online (Coursera)* 2026